Electra Vanguard (ATH-734)

Institutional Alternative Investment Strategy — Quantitative Performance Analysis

Track Record: Jan 2005 – Jun 2026 (258 Months / 21.5 Years) | **Benchmark:** S&P 500 Total Return

ATHÉNÉE INVESTMENT

CAPITAL INTRODUCTION & THIRD-PARTY MARKETING

CAGR (Net)	Ann. Volatility	Sharpe Ratio	Max Drawdown	Alpha (Ann.)	Beta & Corr.
9.28%	**8.89%**	**1.05**	**-15.24%**	**8.68%**	**0.05 | +0.09**
S&P 500: 10.94%	S&P 500: 14.89%	SPX: 0.78 | Sortino: 1.83	SPX: -50.95% | Calmar: 0.61	t-stat: 4.43	Beta t-stat: 1.44

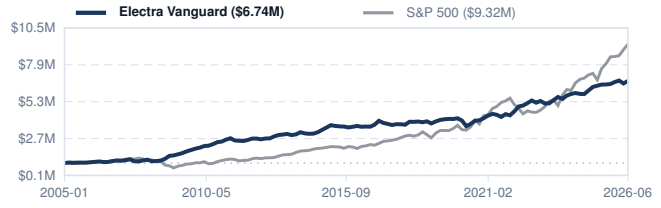
Performance, Risk & Hit Rate Indicators

Indicator	Fund	S&P 500
CAGR (Annualized)	**9.28%**	10.94%
Annualized Mean Return	9.30%	11.54%
Annualized Volatility	8.89%	14.89%
Proxy Sharpe Ratio ($r_f = 0\%$)	**1.05**	0.78
Annualized Sortino Ratio	1.83	1.19
Maximum Drawdown (MDD)	**-15.24%**	-50.95%
Calmar Ratio (CAGR / |MDD|)	0.61	0.21
Cumulative Total Return	574.12%	832.13%
Best / Worst Monthly	+11.65% / -11.95%	+12.82% / -16.80%
Positive Months (Hit Rate)	64.7% (167/258)	67.4% (174/258)
Crisis Hit Rate (Fund > 0 | SPX < 0)	**54.8% (46/84)**	–
Up-Mkt Loss (Fund < 0 | SPX > 0)	30.5% (53/174)	–

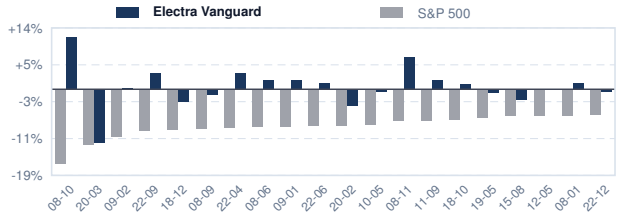
Alpha, Beta, Correlation & Tail Analysis (vs. S&P 500)

Quantitative Metric	Estimate / Diagnostic
Annualized Alpha (α) & t-stat	**8.68% (t=4.43)**
Market Beta (β_{SPX}) & t-stat	0.05 (t=1.44) ($R^2 = 0.8\%$)
Pearson Correlation (r) to S&P 500	**+0.09** (95% Blk CI: [-0.14, 0.34])
Spearman Rank Correlation (ρ)	+0.15 (95% Blk CI: [-0.00, 0.30])
Upside / Downside Capture Ratio	Up: **0.29** | Down: **-0.12**
Cond. Up-Market (SPX > 0)	$\alpha = 5.6\%,\ \beta = 0.14$
Cond. Down-Market (SPX < 0)	$\alpha = -0.2\%,\ \beta = -0.12$
Tail Corr. (SPX $< Q_1$, Worst 25%)	$r = -0.06,\ \rho = -0.03\ (n = 65)$
Tail Corr. (SPX $< D_1$, Worst 10%)	$r = -0.22,\ \rho = -0.15\ (n = 26)$
Worst 20% SPX Months Avg. Return	Fund: +0.60% vs SPX: -5.41%
Cross-Asset Beta (WTI / BTC)	WTI β: 0.05 | BTC β: 0.02
Ljung-Box Serial Correlation	$p_1 = 0.01,\ p_3 = 0.03$ (Serial Corr.)

Cumulative Growth of $1M USD (Fund vs. S&P 500)



Worst 20% S&P 500 Months



Historical Monthly & Yearly Net Return Matrix (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Yearly
2005	0.4%	3.0%	0.7%	-1.9%	0.2%	0.5%	0.1%	0.1%	-0.2%	-0.0%	0.4%	2.5%	**5.8%**
2006	1.6%	0.7%	0.6%	1.9%	-1.9%	-1.6%	0.0%	0.8%	4.3%	1.2%	1.2%	1.7%	**10.7%**
2007	0.2%	-1.3%	-1.0%	4.6%	7.0%	-1.4%	-7.6%	-4.3%	0.2%	-0.4%	6.7%	-0.7%	**1.0%**
2008	1.4%	3.1%	-0.2%	-8.8%	-1.1%	2.1%	2.4%	0.7%	-1.3%	11.7%	7.1%	8.5%	**26.7%**
2009	2.0%	0.3%	0.8%	4.3%	6.6%	-1.4%	5.6%	1.6%	1.8%	3.4%	3.9%	0.8%	**34.0%**
2010	1.3%	2.0%	4.9%	0.5%	-0.5%	1.9%	1.2%	3.5%	3.0%	2.7%	-0.2%	1.9%	**24.5%**
2011	2.2%	3.3%	-0.9%	4.5%	-3.8%	-1.9%	3.0%	-3.7%	2.1%	-2.7%	-1.4%	4.4%	**4.6%**
2012	1.2%	2.5%	1.5%	-1.6%	0.1%	-1.9%	1.7%	0.4%	1.1%	-0.9%	1.3%	3.7%	**9.3%**
2013	3.2%	0.4%	0.4%	0.7%	1.9%	-0.6%	-0.8%	-1.8%	-0.9%	2.7%	2.1%	2.9%	**10.5%**
2014	-1.3%	-1.2%	-1.6%	0.5%	-0.2%	0.6%	-0.7%	4.0%	7.1%	-1.8%	4.2%	0.8%	**10.4%**
2015	5.7%	-1.1%	1.9%	-3.3%	1.8%	-2.6%	2.1%	-2.3%	-0.5%	-1.3%	3.2%	-1.9%	**1.4%**
2016	0.7%	0.9%	-0.8%	-0.8%	-0.0%	0.6%	0.2%	-0.6%	0.3%	3.5%	4.9%	2.4%	**11.8%**
2017	-2.0%	-1.7%	0.3%	-2.2%	-1.6%	-0.5%	2.4%	-0.8%	0.3%	-0.5%	-0.5%	-0.3%	**-6.9%**
2018	6.6%	-0.9%	-2.2%	2.0%	0.5%	0.1%	-1.7%	0.4%	0.6%	1.0%	-1.1%	-2.8%	**2.4%**
2019	3.3%	0.4%	-0.4%	3.1%	-0.7%	1.9%	-1.8%	2.4%	-1.4%	1.3%	-0.3%	1.4%	**9.5%**
2020	-0.1%	-3.6%	-11.9%	2.3%	3.1%	1.3%	4.0%	1.9%	0.8%	-0.2%	0.5%	3.5%	**0.3%**
2021	-0.2%	4.5%	2.3%	0.4%	1.2%	-2.5%	-4.5%	1.1%	3.0%	1.6%	-2.5%	0.6%	**4.7%**
2022	3.6%	2.5%	3.8%	3.6%	0.0%	1.4%	-0.0%	3.4%	3.6%	-0.2%	-2.1%	-0.6%	**20.5%**
2023	0.3%	1.8%	-2.3%	-1.0%	-0.4%	0.5%	2.1%	2.3%	2.5%	1.4%	-0.3%	-2.1%	**4.9%**
2024	3.0%	1.2%	2.0%	0.1%	0.6%	0.4%	0.3%	-1.3%	0.7%	-1.1%	2.4%	2.0%	**10.8%**
2025	2.9%	1.1%	1.2%	0.1%	0.1%	1.1%	-0.9%	1.0%	-0.0%	0.5%	0.6%	1.7%	**9.7%**
2026	1.7%	-0.2%	-6.1%	3.4%	2.2%	0.3%	-	-	-	-	-	-	**1.0%**

Fund Terms & Commercial Overview

Vehicle:	**Commingled Fund**	**Return Basis:**	**Net of Fees**	**Fee Structure:**	**2% Mgmt / 20% Perf**	**Liquidity:**	**Quarter Liquidity**
Lock-Up Period:	**No Lockup**	**Min. Investment:**	**1M USD**	**Current AUM:**	**250M**	**Domicile:**	**Cayman Islands**

Risk Disclosures, Legal Notice & Quantitative Appendix

Fund Reference: Electra Vanguard (ATH-734) (Jan 2005 - Jun 2026 (258 Months / 21.5 Years))

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Appendix A: Extended Statistical, Bootstrap & Factor Diagnostics

Diagnostic Family	Methodology / Statistical Specification	Empirical Value
Pearson Correlation CI	95% Confidence Interval (IID Bootstrap vs. Moving Block $B = 3$, 1,000 iters)	Std: [-0.11, 0.30] | Blk: [-0.14, 0.34]
Spearman Rank Corr. CI	95% Confidence Interval (Rank Monotonic Co-Movement, IID vs. Block $B = 3$)	Std: [0.03, 0.28] | Blk: [-0.00, 0.30]
Worst-Case Co-Dependence	97.5th Percentile Upper-Bound Block Bootstrap Correlation (PSD Projection)	$\rho_{\text{wc}} = 0.34$
Serial Correlation (Lag 1 & 3)	Ljung-Box Q-Statistic p-values testing for return smoothing / illiquidity bias	$p_1 = 0.01,\ p_3 = 0.03$ (Serial Corr.)
Conditional Up-Market OLS	Annualized Alpha & Beta estimated during positive S&P 500 months ($n = 174$)	$\alpha = 5.6\%,\ \beta = 0.14$ ($t_\alpha = 1.74$)
Conditional Down-Market OLS	Annualized Alpha & Beta estimated during negative S&P 500 months ($n = 84$)	$\alpha = -0.2\%,\ \beta = -0.12$ ($t_\alpha = -0.03$)
Commodity Factor (WTI)	OLS Regression against Front-Month WTI Crude Oil Monthly Returns	$\beta = 0.05$ ($t = 3.13$), $\alpha = 7.2\%$
Digital Asset Factor (BTC)	OLS Regression against Bitcoin (BTC-USD) Monthly Returns	$\beta = 0.02$ ($t = 1.64$), $\alpha = 6.9\%$

Appendix B: Quantitative Framework & Indicator Definitions

- **CAGR & Annualized Volatility:** Compound Annual Growth Rate is computed geometrically as CAGR $= \left(\prod_{t=1}^{T}(1 + r_t)\right)^{12/T} - 1$. Annualized Volatility is sample standard deviation scaled by $\sqrt{12}$: $\sigma_{\text{ann}} = \hat{\sigma}_m\sqrt{12}$. Proxy Sharpe Ratio assumes $r_f = 0\%$ ($\bar{r}_{\text{ann}}/\sigma_{\text{ann}}$).
- **Maximum Drawdown (MDD) & Calmar Ratio:** Maximum peak-to-trough decline of cumulative compounded wealth $W_t = \prod_{s\le t}(1 + r_s)$, defined as MDD $= \min_t\left((W_t - \max_{s\le t} W_s)/\max_{s\le t} W_s\right)$. Calmar Ratio is CAGR/|MDD|.
- **OLS Alpha (α), Beta (β) & Capture Ratios:** Estimated via Ordinary Least Squares $r_{\text{fund},t} = \alpha_m + \beta r_{\text{SPX},t} + \varepsilon_t$, where $\alpha_{\text{ann}} = 12 \times \alpha_m$. Upside (Downside) Capture measures the ratio of the fund's mean return to the S&P 500's mean return during positive (negative) S&P 500 months.
- **Tail Correlations & Moving Block Bootstrap:** Tail correlations evaluate linear (r) and rank (ρ) co-movement during S&P 500 drawdowns below the 25th percentile (Q_1) and 10th percentile (D_1). Moving Block Bootstrap ($B = 3$ months) preserves local serial dependencies when constructing 95% confidence intervals.

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